Churn Risk and Retention Analysis Report

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Findings

Our analysis of churn risk using Accelerated Failure Time (AFT) models revealed consistent results across the Log-Logistic, Log-Normal, and Weibull distributions. All models identified strong predictors of churn. Specifically, customers in the **E-service**, **Plus service**, and **Total service** categories have significantly higher survival probabilities. For example, the Log-Logistic model shows exp(coef): 2.84 (E-service), 2.39 (Plus service), and 3.33 (Total service), suggesting these customers are much less likely to churn compared to the baseline group.

Other key factors influencing churn include:

- **Age:** Older customers are less likely to churn (exp(coef): 1.03).
- **Marital status:** Unmarried users have a higher churn risk (exp(coef): 0.65), indicating marriage is protective.
- **Internet usage:** Internet users show a significant risk of churn (exp(coef: 0.44)), highlighting this segment as vulnerable.

Survival Function Plot

Valuable Segments

The most valuable segments are defined as customers with both high predicted Customer Lifetime Value (CLV) and low churn risk. Based on our analysis, **Plus service** and **E-service** groups stand out:

- **Plus service:** Mean CLV $\approx$ 2798 units.
- **E-service:** Mean CLV $\approx$ 2553 units.

These segments also have large customer counts (217–281), making them critical for revenue retention. In this context, being "valuable" means having high revenue potential *and* a lower probability of churn, maximizing return per customer.

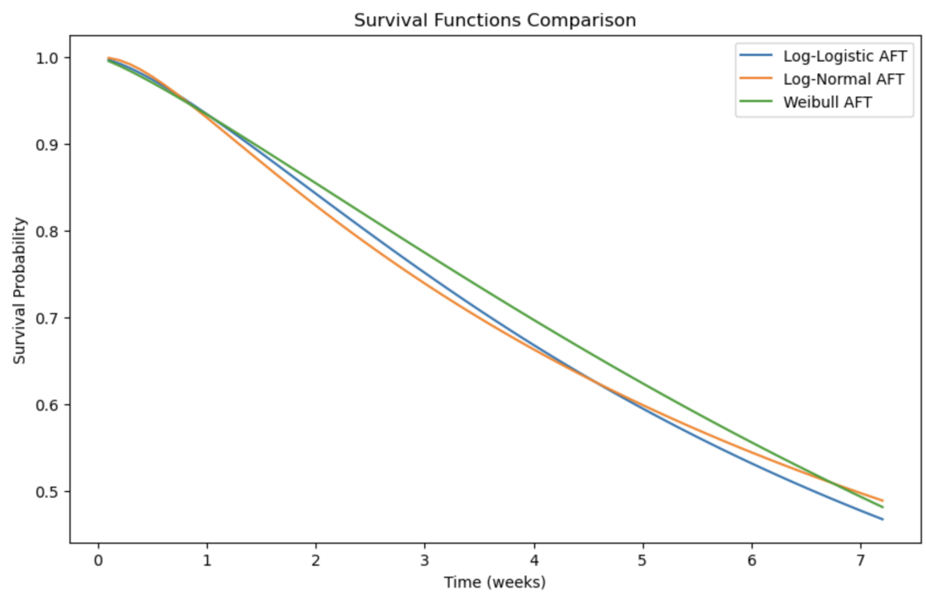

Figure 1: Survival function comparison across AFT models: Log-Logistic, Log-Normal, and Weibull. The plot illustrates the probability of customer retention over time.

Retention Budget Estimate

Assuming the dataset represents the entire customer population, and with an average CLV of around 2000 units across key segments, a conservative annual retention budget should be approximately **5–10% of total CLV**. For a customer base of around 1000 individuals, this equates to a retention budget in the range of **100,000 to 200,000 units per year**.

Furthermore, early identification of at-risk customers—such as internet users or unmarried subscribers—allows more precise targeting of retention efforts. We estimate that around 25–30% of customers fall into high-risk categories based on survival probabilities within the first year.

Recommendations

To strengthen retention, we recommend:

- Implementing **personalized retention campaigns** focused on high-risk users.
- Establishing **loyalty programs** tailored to the most valuable segments (E-service and Plus service).
- Enhancing **monitoring of engagement metrics** (e.g., usage frequency, feature adoption) to proactively address churn risks.